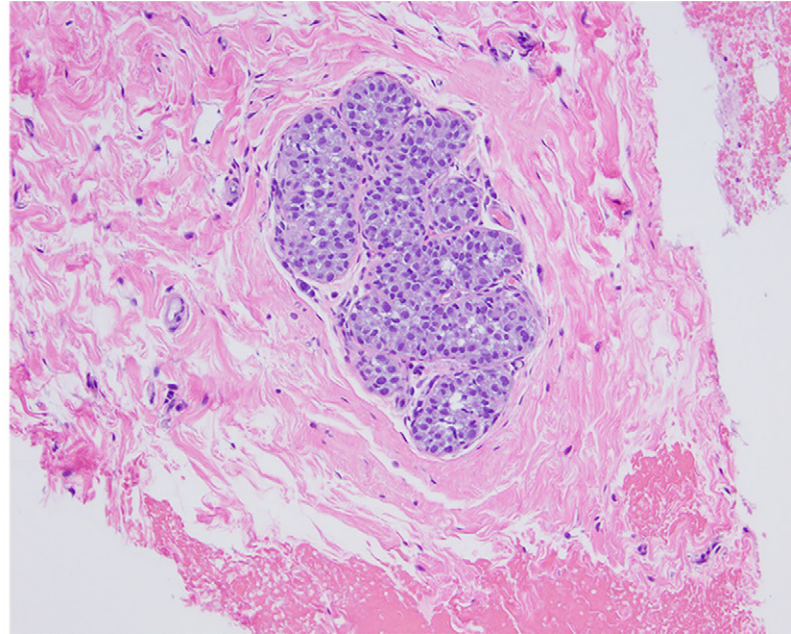
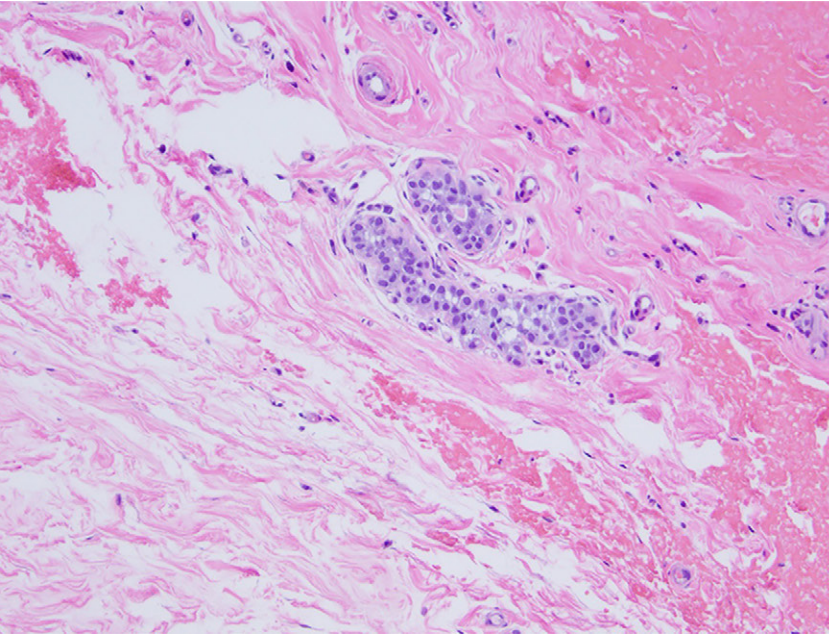


Update on Lobular Neoplasia

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Lobular neoplasia (LN) is a histopathologic entity that encompasses both lobular carcinoma in situ (LCIS) and atypical lobular hyperplasia (ALH). Management of LN is known to be variable and institutionally dependent. The variability in approach after a diagnosis of LN at percutaneous breast biopsy derives in part from heterogeneity in the literature, resulting in a range of reported upgrade rates to malignancy after initial identification at percutaneous biopsy, and also from historical shifts in understanding of the natural history of LN. It has become increasingly recognized that not all LN is the same and that distinct variants of LN such as pleomorphic LCIS and florid LCIS have distinct natural histories and distinct likelihoods of upgrade to malignancy. In addition, it is also increasingly understood that appropriate management of LN relies on scrupulous radiologic-pathologic correlation. This review details the imaging features and histopathologic nature of ALH, classic-type LCIS, and the LCIS variants; addresses changes in the historical understanding of this entity contributing to confusion regarding its management; and discusses the importance of performing radiologic-pathologic correlation after percutaneous biopsy to help guide appropriate management steps when LN is encountered. In addition to the short-term implications of an LN diagnosis in terms of upgrade and surgical outcomes, the long-term implications of an LN diagnosis regarding risk of developing a later breast cancer are examined.

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Introduction

Lobular neoplasia (LN) is a histopathologic entity that encompasses both lobular carcinoma in situ (LCIS) and atypical lobular hyperplasia (ALH). Management of LN is known to be variable and institutionally dependent. A 2019 survey of academic imaging centers in the United States found that 71% of respondents thought that LCIS should be excised and 61% thought that ALH should be excised. However, a sizable minority opted for surveillance of both (1). Another case-based scenario survey from Canada also demonstrated simi-

lar variability in recommendations for surveillance, risk-reduction management, and surgical excision after a classic LCIS diagnosis (2).

The variability in management approaches after LN is identified at percutaneous breast biopsy has several causes. First, the literature evaluating upgrade to either in situ or invasive cancer in the setting of an LN percutaneous biopsy result is heterogeneous. Studies are often small, single-institution, and performed with variable biopsy devices and varying modalities and end points, for example surgery versus



Quiz questions for this article are available through the Online Learning Center.

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Abbreviations: ALH = atypical lobular hyperplasia, H-E = hematoxylin-eosin, ILC = invasive lobular carcinoma, LCIS = lobular carcinoma in situ, LN = lobular neoplasia

TEACHING POINTS

- Pathologists have long recognized the difficulty and potential for subjectivity in distinguishing between LCIS and ALH at pathologic analysis of biopsy specimens; as has been stated, “The histological distinction between ALH and LCIS is often problematic and based largely on subtle differences to which the lobules are involved.”
- In contrast to classic LCIS, the variants are more likely to be nonincidental findings, meaning that the pathologic entity is associated with the imaging-targeted finding (most commonly calcifications at mammography).
- If a result of LN does not explain the imaging finding or match the level of suspicion, the result should be termed *discordant*, and this should be recorded as an addendum to the biopsy report. Further tissue should be obtained with repeat biopsy or surgical excision.
- A diagnosis of LN is associated with a greater than average lifetime risk of developing breast cancer.
- Recommendations for supplemental MRI screening in women with a history of LN are based more on lifetime risk observations, extrapolations from the high-risk patient literature, and recognition that women may have multiple simultaneous risk factors.

observation. This makes it difficult to assess upgrade rates to malignancy robustly. In addition, not all LN is the same, and there is increasing recognition that distinct variants of LN have distinct natural histories and distinct likelihoods of upgrade to malignancy. Recent meta-analyses, systematic reviews, and studies with stringent radiologic-pathologic correlation as well as improved understanding of the variant subtypes are changing understanding of LN and its attendant management.

This review details the imaging features and histopathologic nature of ALH, classic-type LCIS, and the LCIS variants; addresses changes in historical understanding of this entity contributing to confusion regarding its management; and discusses the importance of performing radiologic-pathologic correlation after percutaneous biopsy to help guide appropriate management steps when LN is encountered. In addition to the short-term implications of an LN diagnosis in terms of upgrade and surgical outcomes, the long-term implications of an LN diagnosis regarding risk of developing a later breast cancer are examined.

Historical Shifts in Understanding LN

Historical changes in understanding of LN have contributed to confusion regarding its management. In 1919, Ewing (3) referred to LN as an “atypical proliferation of acinar cells.” In 1941, Foote and Stewart (4) described LCIS as an invasive cancer precursor for which mastectomy was considered to be appropriate. Then in 1978, Haagensen et al (5) observed that LCIS had only low rates of progression to cancer. The authors opined that LN was a risk marker for later cancer development but not a likely precursor of malignancy, and therefore

recommended surveillance with clinical breast palpation to be performed every 4 months instead of mastectomy.

In 1992, Eusebi et al (6) realized that not all LCIS was the same. Through their description and recognition of the variant pleomorphic LCIS, Eusebi et al (6) opened the door to understanding that LCIS is a heterogeneous entity and that distinct management strategies for distinct types of LN might be appropriate. Today, LN is understood to be both a risk marker for later development of breast cancer and, in some forms, a nonobligate precursor. This recognition has further shaped management approaches.

Evidence That LN Is a Nonobligate Cancer Precursor

How do we know that LN may sometimes lead to cancer? In women with a history of LN, there is a greater lifetime risk of cancer (7,8). Although future malignancy may be ductal or lobular and may occur in either breast, later cancers are more likely to occur in the ipsilateral breast and are more likely to be lobular (9–11).

In addition, genetic and molecular studies have demonstrated that LN and invasive lobular carcinoma (ILC) are likely to be found in association with each other and are phenotypically and genetically similar. Both usually demonstrate loss of chromosome 1q, recurrent deletions of 16q, and loss of E-cadherin (due to mutations of the *CDH1* gene on 16q) (12). Shared genetic changes and clonal relationships are frequently observed among associated LN and ILC (13–15). One study that evaluated mastectomy specimens in 50 cases with copy number profiling and whole exome sequencing demonstrated high numbers of clonal relationships between LCIS and ILC in the same specimen. Specifically, the authors found concordant gene copy number changes and identical mutations in LCIS and ILC within the same patient, again suggesting that LCIS serves as a precursor lesion.

LN Types and Upgrade Rates

It is critical to understand that LN is a heterogeneous disease. Appropriate management—in practice deciding between surveillance versus surgical referral or excision—has been institutionally variable and controversial based on a wide range of reported upgrade rates to either in situ or invasive malignancy of 0%–43% for excised LCIS and 0%–67% for excised ALH (16). However, once it is understood that there are variant subtypes of LN with distinct natural histories and that certain types of LN are more likely to be associated with malignancy, the broad upgrade rate ranges can be broken down and management approaches can be more evidence based. With this in mind, it is important to delve more deeply into the imaging and histopathologic features of LN.

LN (ALH or Classic LCIS)

LN is an atypical proliferation of noncohesive monomorphic cells in the terminal duct lobular unit (TDLU). The cells are noncohesive because of the loss of E-cadherin, which is an adhesion protein. Two cell types may be seen: A cells, which are small with minimal cytoplasm and homogeneous nuclei; and B cells, which are larger than A cells and more varied in size and

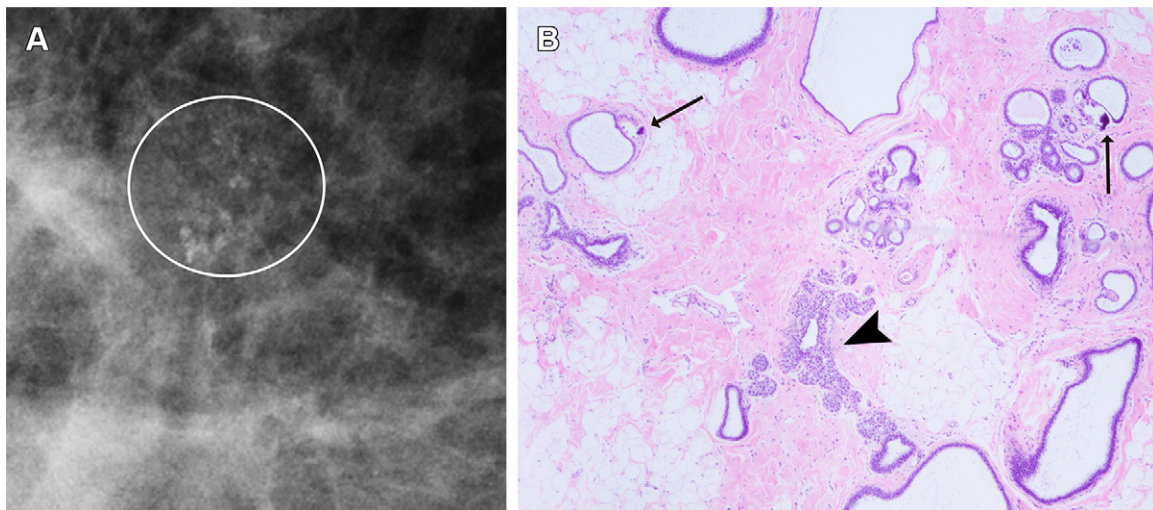


Figure 1. Incidental LCIS in a 43-year-old woman who presented for recall after right breast calcifications were identified at annual screening mammography. (A) Magnification craniocaudal view shows grouped amorphous calcifications (white circle). The calcifications were biopsied under stereotactic guidance, yielding incidental classic LCIS and benign cysts and ductules with associated benign microcalcifications. (B) Low-power photomicrograph shows calcification correlating with the mammographic findings with associated benign cysts and ductules (arrows); the calcifications explain the targeted calcifications seen at mammography. The LCIS (arrowhead) is not associated with calcifications and is incidental to the biopsy target. (Hematoxylin-eosin [H-E] stain.)

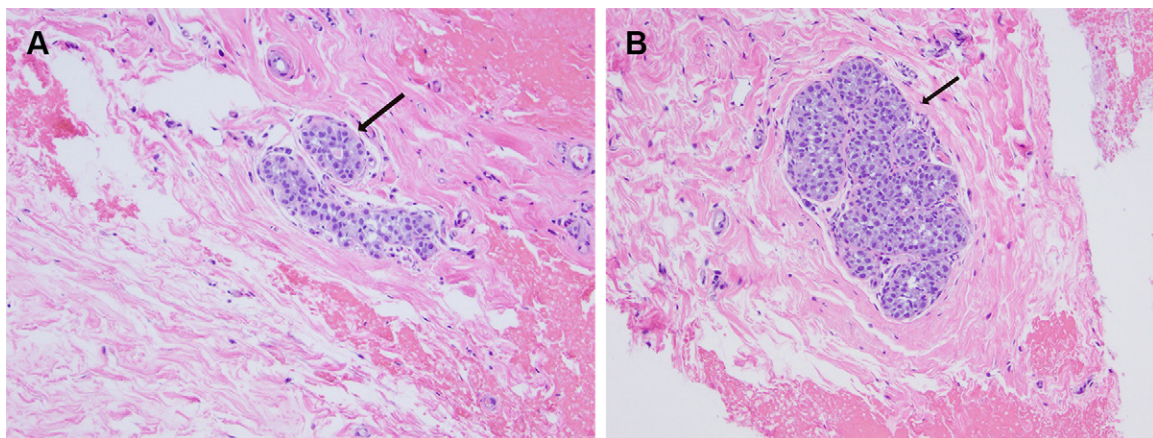


Figure 2. The distinction between ALH and classic LCIS may be somewhat subjective, as has been noted in the pathology literature. (A) High-power photomicrograph shows ALH (arrow). The lobular unit in the image demonstrates acini that are filled with cells, but there is no expansion. (H-E stain.) (B) In contrast, high-power photomicrograph shows classic LCIS, in which cells both fill and expand the acini (arrow). (H-E stain.) The number of lobules does not matter in terms of making the distinction between ALH and classic LCIS, but the amount of proliferation within the lobular unit does.

shape, contain relatively more cytoplasm, and demonstrate vesicular chromatin (17). Cytoplasmic vacuoles may be seen with signet ring morphology.

LN is more commonly found in premenopausal women and is often bilateral and most commonly asymptomatic. Nonvariant LN is most often incidental at imaging, meaning that it was not the imaging-targeted entity but that it just happened to be seen by the pathologist in histopathologic samples (Fig 1) (17). However, it may rarely (<2% of the time) be associated with imaging findings, including calcifications (18,19).

ALH and classic LCIS would not be expected to appear as mass lesions, although they may be found in association with mass lesions such as fibroadenomas. They may also be associated with other benign lesions like sclerosing adenosis and other high-risk lesions such as papillary lesions or radial scars

(10). LN has been identified in 0.5%–3.6% of core biopsy specimens and in 0.04%–1.2% of breast reduction tissue specimens, but its true natural incidence is unknown precisely because it is often undetected (19). However, the incidence is noted to have increased in the 1980s, especially in women over age 50 years, a finding attributed in large part to the rise of screening (20).

ALH versus LCIS

LN encompasses both classic-type LCIS and ALH. In LCIS over half of the acini of the TDLU are filled and expanded with atypical cells, while in ALH less than half of the acini are filled and expanded (Fig 2). Pathologists have long recognized the difficulty and potential for subjectivity in distinguishing between LCIS and ALH at pathologic analysis of biopsy specimens; as has been stated, “The histological distinction between

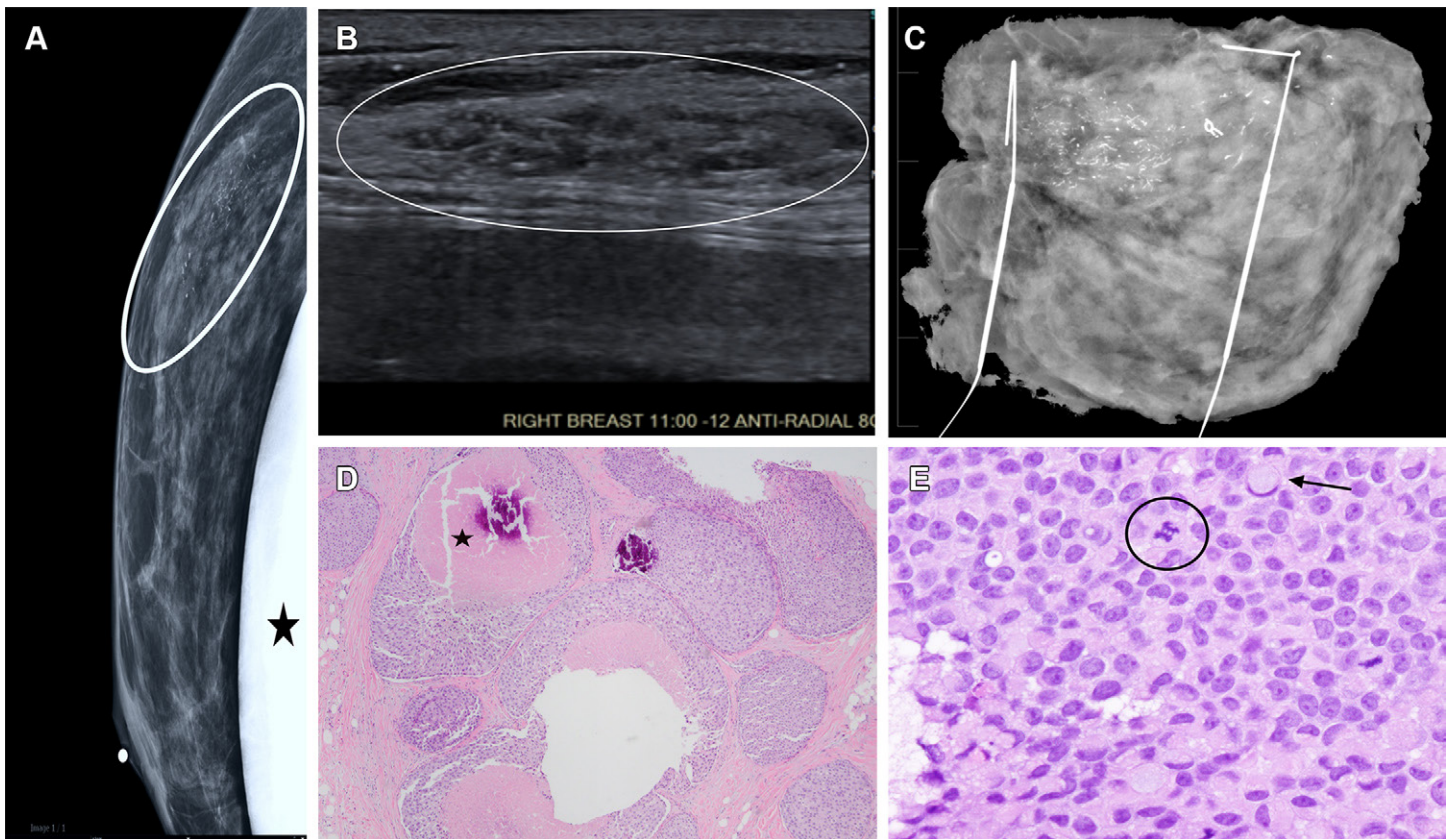


Figure 3. Pleomorphic LCIS in a 66-year-old woman recalled from screening mammography for right breast calcifications. (A) Right lateral implant-displaced mammographic view shows suspicious segmental fine pleomorphic calcifications (white oval). Note that the patient also has a silicone implant (★). (B) Image from US performed for further evaluation shows a hypoechoic mass with hyperechoic foci (white oval), correlating with the calcifications seen at mammography. The silicone implant is partially visualized. US-guided core biopsy was performed. Calcifications were seen at radiography of the core samples (not shown). Histopathologic analysis of the biopsy specimen showed pleomorphic LCIS with associated calcifications, and the patient was sent for surgical excision. (C) Specimen radiograph after bracketed wire localization and lumpectomy shows the distal aspect of two wires, the biopsy marker clip, and the calcifications. (D) Low-power photomicrograph from the surgical excision shows pleomorphic LCIS with comedo necrosis (★), correlating with the calcifications seen at radiography. (H-E stain.) (E) High-power photomicrograph shows a representative mitotic figure (black circle), consistent with the proliferative nature of this lesion. Signet ring cells with cytoplasmic mucin vacuoles are also present (arrow). In contrast to classic LCIS, pleomorphic LCIS demonstrates variability in nuclear size, a sign of underlying genetic instability. (H-E stain.)

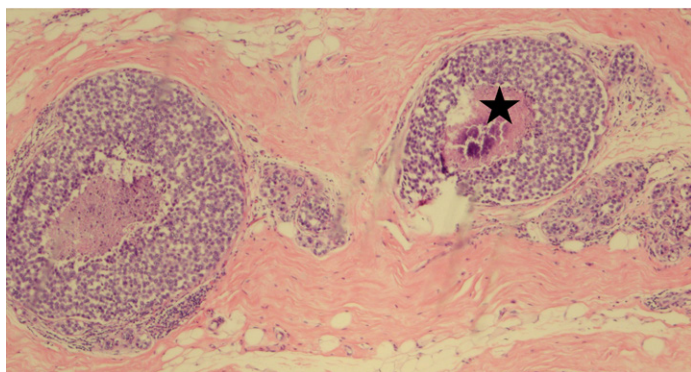


Figure 4. Histopathologic analysis of core biopsy specimen from a 76-year-old woman with pleomorphic LCIS. Low-power photomicrograph shows typical central necrosis and calcifications (★). (H-E stain.)

ALH and LCIS is often problematic and based largely on subtle differences to which the lobules are involved” (21). For this reason, some pathologists prefer to use the term *lobular neoplasia* rather than make a distinction between ALH and LCIS, especially in core biopsy specimens alone. When well sampled, concordant, and pure (ie, without associated other lesions such

as atypical ductal hyperplasia [ADH] that might raise the likelihood of upgrade to malignancy), ALH and classic LCIS are unlikely to be upgraded to malignancy.

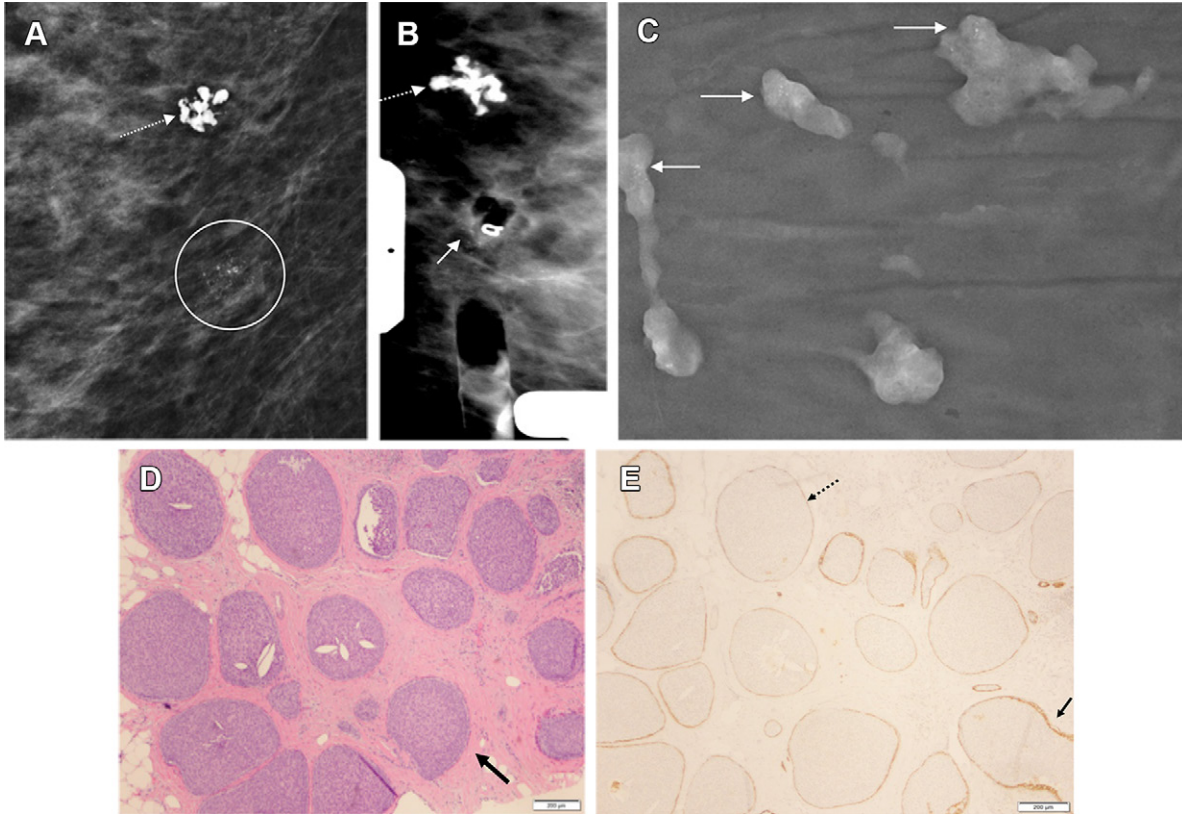
The Variants (Nonclassic LN)

The LN variants are relatively rare and, per the World Health Organization (WHO) classification, include pleomorphic LCIS and florid LCIS. The latter was added in 2019. LCIS described as necrotic should also be placed in the variant category.

In contrast to classic LCIS and ALH, the variants are more commonly found in postmenopausal women (10,22). In a study that evaluated 179 women with pleomorphic LCIS, the mean age at diagnosis was 60 years (23). In contrast to classic LCIS, the variants are more likely to be nonincidental findings, meaning that the pathologic entity is associated with the imaging-targeted finding (most commonly calcifications at mammography) (Figs 3–5) (10,24).

The variants are also more likely to be upgraded to and associated with malignancy. Upgrade rates to malignancy for pleomorphic LCIS and florid LCIS are reported to be 25%–60% (25). Classic LCIS and the variants may be found together in the same biopsy or surgical specimen (17).

Figure 5. Florid LCIS in a 62-year-old woman who initially presented for screening mammography with a finding of new right breast calcifications. (A) Diagnostic lateral magnification mammographic view shows suspicious grouped pleomorphic calcifications (white circle). Coarse popcorn calcifications from benign hyalinized fibroadenoma (arrow) are incidentally noted, superior to the suspicious calcifications. Stereotactic biopsy was performed. (B) Post-clip placement stereo mammographic image shows the marker clip in good position with residual calcifications (solid arrow). Popcorn calcifications are again seen (dotted arrow). (C) Specimen radiograph shows adequate sampling of the calcifications (arrows). (D) Low-power photomicrograph shows the typical marked distention of the acini (arrow), which are filled with fairly bland cells characteristic of florid LCIS. (H-E stain.) (E) Low-power photomicrograph (E-cadherin immunohistochemical stain) shows brown staining from the myoepithelial cells, with E-cadherin surrounding the acini (solid arrow). However, many of the cells are E-cadherin negative (dotted arrow), typical for lobular lesions.



Pleomorphic LCIS.—At histopathologic analysis, pleomorphic LCIS is composed of large pleomorphic discohesive cells with marked nuclear atypia, akin to the high-grade nuclei seen in ductal carcinoma in situ (DCIS) (26). Signet-ring cell morphology may be seen, as in classic LCIS (Fig 3). High-grade cytomorphology, massive acinar distention, comedo necrosis, and microcalcification are commonly seen (Figs 3, 4), although necrosis and calcification are not mandated for diagnosis (17). Loss of E-cadherin expression is typical. While classic LCIS is usually estrogen receptor (ER) and progesterone receptor (PR) positive and human epidermal growth factor receptor 2 (HER2) negative, pleomorphic LCIS is more varied in its receptor profile and while typically HER2 negative may also be HER2 positive (19,27).

At imaging, pleomorphic LCIS most commonly manifests as calcifications at mammography, but it may also manifest as a mass lesion or distortion and may be noncalcified. A retrospective review of pleomorphic LCIS found that of 16 cases of nonupgraded pleomorphic LCIS, 13 (81%) manifested as fine pleomorphic calcifications, one (6%) as distortion with calcifications, one (6%) as a mass lesion, and one (6%) as nonmass enhancement at MRI. The median imaging size was 11 mm, with a mean of 14 mm and range of 3–47 mm (24). Another

study that reported on 37 cases of pleomorphic LCIS found that 31 cases (84%) manifested as calcifications, although further descriptors were not reported (24,28).

Florid LCIS.—Florid LCIS demonstrates nuclear features akin to those of classic LCIS. Like classic LCIS, florid LCIS may have type A and/or type B cells. However, the architectural features of florid LCIS are distinct from those of classic LCIS, with mass-forming marked distention of the acini or ducts, which are filled with fairly bland cells and very little stroma (Fig 5) (25). Importantly, florid LCIS is not synonymous with *extensive LCIS*, a term denoting widespread terminal duct lobular unit (TDLU) involvement without acinar distention (19).

As with pleomorphic LCIS, comedo necrosis and calcifications are commonly present. Like pleomorphic LCIS, florid LCIS is typically associated with calcifications at imaging, but it may also appear as a mass lesion or distortion and may be noncalcified. As with classic LCIS and pleomorphic LCIS, loss of E-cadherin expression is typical. Florid LCIS is typically ER and PR positive, although it may also be HER2 positive.

Necrotic LCIS.—Both pleomorphic LCIS and florid LCIS may demonstrate comedo necrosis, meaning that expansive

necrosis will be seen centrally in the involved duct space at histopathologic analysis. In contrast, classic LCIS will not demonstrate comedo necrosis, although small foci of necrosis may be identified (29). Confusingly, some authors and pathologists use the term *necrotic LCIS* synonymously with *florid LCIS*, while others may use the term *necrotic* as a descriptor to convey the comedo necrosis seen at histopathologic analysis in both pleomorphic LCIS and florid LCIS. While necrotic LCIS is not an official World Health Organization (WHO) variant lesion, the term may still be used in the literature (30). An LCIS lesion described as exhibiting comedo necrosis should be considered and managed as a variant (29).

Management of LCIS

To determine appropriate management of LN, the histopathologic type of LN must be identified and radiologic-pathologic correlation with assessment of any discordance must be performed. The high upgrade rates for pleomorphic LCIS and florid LCIS lie behind recommendations to treat these lesions with surgical excision (26). The WHO Classification of Tumours (fifth edition) specifically recommends excision for both pleomorphic LCIS and florid LCIS (10).

However, it should be recognized that in 2018, the eighth edition of the American Joint Committee on Cancer TNM staging revised its guidelines for LCIS. In this edition, LCIS was removed from the staging system and considered benign. There was debate about whether to include pleomorphic LCIS in the staging system as pTis, but the committee decided that there were insufficient outcomes data to do so (26,31). A discussion of treatment of the variants is beyond the scope of this review.

Radiologic-Pathologic Correlation and Evaluation for Discordance

As stated in the American College of Radiology (ACR) practice parameters for breast biopsies, the radiologist must evaluate biopsy results for discordance and thereafter make an appropriate management recommendation (32,33). Assessing discordance and sampling entails review of the nature and level of suspicion of the imaging finding and assessment of appropriate sampling of the targeted lesion. In general, the needle gauge used, number of cores, and use of vacuum-assisted devices have been shown to impact upgrade rates. Smaller needle gauge and nonvacuum technique are more likely to lead to upgrade owing to greater chance of undersampling (34), although not all studies have demonstrated such an association for LN specifically (35,36).

Regardless of the needle gauge or device used, the histopathologic result must be evaluated to assess whether it explains the imaging target. A histopathologic result of ALH alone would not appropriately explain a mass lesion seen at US (Fig 6). If a result of LN does not explain the imaging finding or match the level of suspicion, the result should be termed *discordant*, and this should be recorded as an addendum to the biopsy report. Further tissue should be obtained with repeat biopsy or surgical excision (Fig 7). In general, a multidisciplinary team approach including discussion as to histopathologic findings with the pa-

thologist is critical and will allow the most optimal evaluation of concordance and recommendations for management.

Evidence from the Literature

The importance of identifying LCIS variants and performing radiologic-pathologic correlation for assessment of concordance in determining appropriate LN management has been increasingly recognized. Multiple studies have shown that incidental classic LCIS and ALH have very low upgrade rates (30,37–39). In contrast to classic LCIS and ALH, discordant lesions and the variants have higher upgrade rates, warranting excision (30,37,39).

A recent systematic review and meta-analysis that evaluated LN upgrade risk to malignancy illustrates this (16). The authors included articles published from 1998 to 2020 that reported rates of upgrade to malignancy for classic LN, identifying 65 studies. They determined which studies reported on exclusion of variant LCIS, imaging-discordant lesions, and outcomes for at least 70% of lesions. A separate analysis was then performed for the 16 studies that met all three criteria, and pooled upgrade rates were estimated using random-effects models.

Upgrade rates in all 65 studies included in the systematic review were 0%–67% for excised ALH and 0%–43% for excised LCIS (16). However, the overall pooled upgrade rate to invasive or in situ disease in the 16 studies was only 3.1% (95% CI: 1.8, 5.2). The upgrade rate to invasive disease alone was 1.3% (95% CI: 0.7, 2.4). Isolated ALH demonstrated a pooled upgrade rate to all malignancy of 2.5% (95% CI: 1.6, 3.9) and to invasive disease alone of 0.4% (95% CI: 0.0, 4.2). LCIS demonstrated a pooled upgrade rate to any malignancy of 5.8% (95% CI: 2.9, 11.3) and to invasive disease alone of 3.5% (95% CI: 2.0, 5.9).

The optimal approach for assessing the literature and for illustrating the impact of a radiologic-pathologic-sensitive approach can be demonstrated by focusing on a representative individual study. In a 2008 retrospective single-institution study by Hwang and colleagues (30), upgrade rates of LN after multimodality core biopsy were assessed with attention to variants and discordance. There were 48 cases of ALH, with an upgrade rate to malignancy of only 2% (one of 48 cases). However, nine of 39 cases (23%) of pure LCIS were upgraded, a high percentage, suggesting that LCIS would not be appropriate for surveillance.

However, the authors also categorized LCIS cases by subtype. After excluding the variants (described as pleomorphic LCIS and LCIS with necrosis in this particular study), classic LCIS was found to have a much lower (albeit still high) upgrade rate of 12% (three of 26 cases). On the other hand, pleomorphic LCIS and LCIS with necrosis demonstrated persistent high upgrade rates of 43% (three of seven cases) and 50% (three of six cases), respectively.

An additional level of review was then applied, evaluating cases for discordance. When the authors removed discordant cases (all three classic LCIS cases, two of three pleomorphic LCIS cases, and one of three necrotic LCIS cases), upgrade rates for LN variants remained high (20% [one of five cases] for pleomorphic LCIS and 40% [two of five cases] for necrotic LCIS). However, the classic LCIS upgrade rate for nonvariant

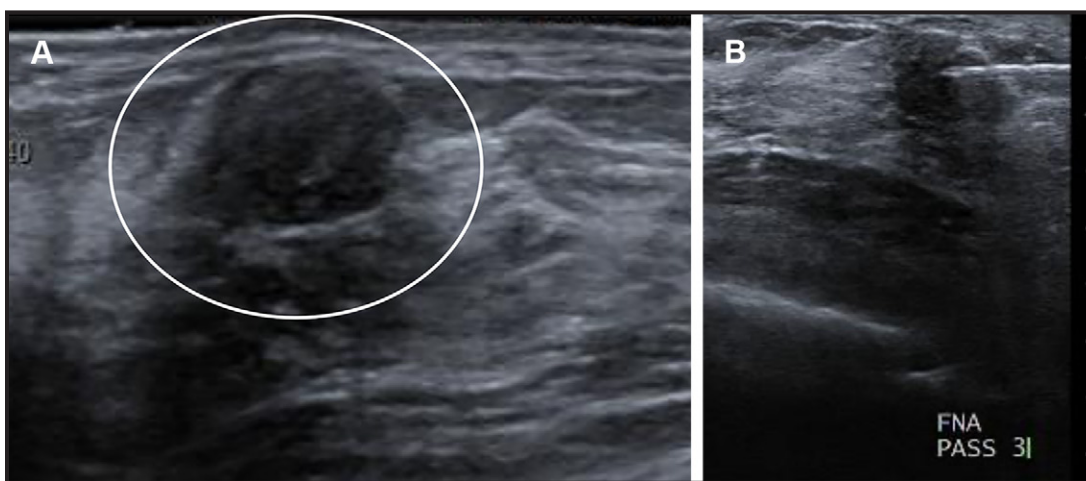


Figure 6. Fibroadenoma with associated ALH in a 42-year-old woman with a palpable breast lump. Diagnostic mammograms (not shown) were normal. (A) Transverse gray-scale US image shows an oval hypoechoic parallel mass (white circle). (B) Image from US-guided fine-needle aspiration, which yielded ALH, discordant with the targeted mass lesion. The patient was sent for surgical excision with the finding of a fibroadenoma with associated ALH, now explaining the imaging finding.

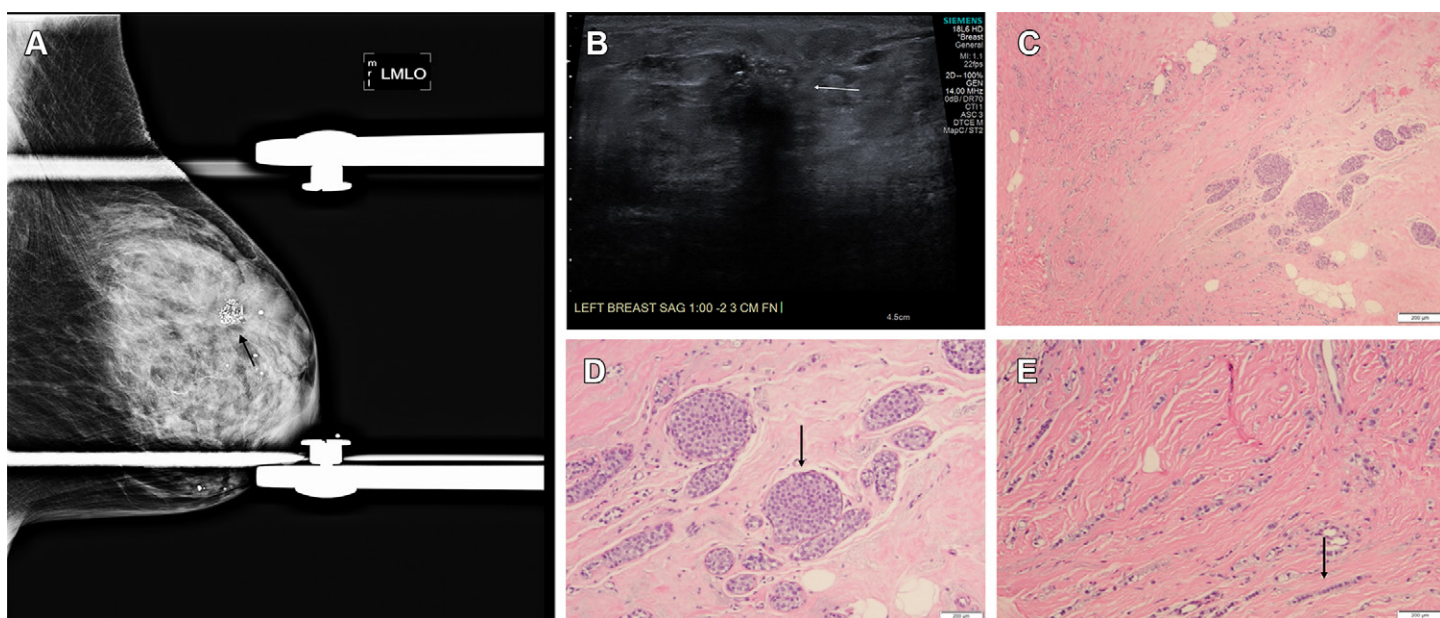


Figure 7. Classic LCIS in association with ILC in an 85-year-old woman with a palpable breast lump. (A) Diagnostic left mediolateral oblique spot compression mammogram shows grouped coarse heterogeneous calcifications and associated distortion (arrow), correlating with the BB marker at the site of the palpable area. (B) Image from same-day US shows calcifications with an associated hypoechoic irregular mass (arrow), correlating with the mammographic finding and with the palpable region of concern. Initial biopsy performed under US guidance demonstrated classic LCIS in a background of dense stromal fibrosis, a result termed *discordant* given the imaging findings of suspicious calcification and a mass. Surgical excision was recommended. (C) Low-power photomicrograph of the surgical excision specimen shows adjacent but distinct classic LCIS (right side of the slide) in association with ILC (left side of the slide). (H-E stain.) (D, E) High-power photomicrographs of the same area show classic LCIS with typical small homogeneous cells (arrow in D), while (E) shows adjacent ILC with single-cell infiltration and no gland formation (arrow).

and concordant cases dropped to 0%. This 0% upgrade rate justifies surveillance rather than excision.

In comparison, other studies that do not perform radiologic-pathologic concordance, do not separate out variants consistently, or include categories of LCIS not otherwise specified will be more likely to report high upgrade rates (40). This demonstrates the importance of complete analysis and radiologic-pathologic correlation, both when reviewing one's own results and when searching the literature to assess upgrade numbers for LN.

Nonincidental versus Incidental LCIS

As mentioned earlier, classic LCIS may occasionally be associated with targeted calcifications; in this case, the LCIS would

be a nonincidental targeted finding. Very few studies in the literature specifically assess the upgrade rates of nonincidental calcified classic LCIS. Even reviews that carefully maintain assessment of discordance and exclude variants in analysis do not necessarily make a distinction between incidental noncalcified classic LCIS and nonincidental calcified LCIS in analysis (16,30).

A recent study that assessed upgrade rates of calcified versus noncalcified LCIS evaluated 81 cases of LN diagnosed at core needle biopsy (41). Sixty-four biopsies yielded concordant incidental LN, not associated with the calcifications. Of these, 42 underwent excision with an upgrade of one case (2.4%) (this case was described as an incidental ILC, remote from the biopsy site). The calcifications in these incidental noncalcified

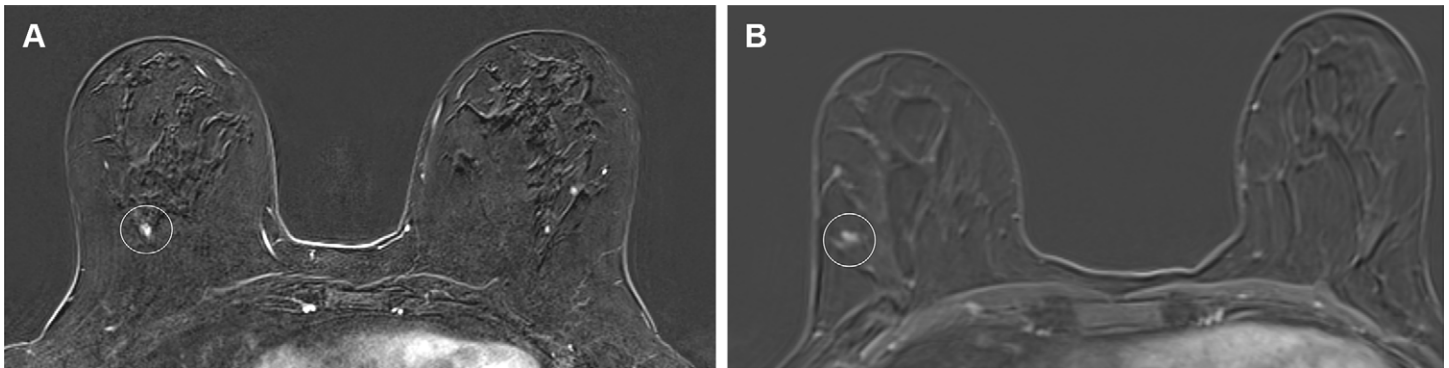


Figure 8. Two cases of LN at MRI demonstrate the difficulty of predicting upgrade in this setting. **(A)** Benign fibroadenoma and classic LCIS in a 53-year-old patient with a high-risk family history. Axial contrast-enhanced subtraction T1-weighted image from annual screening MRI shows a 0.5-cm enhancing mass (white circle) with a type 1 kinetic curve, suspicious. MRI-guided biopsy yielded classic LCIS. The lesion yielded benign fibroadenoma and classic LCIS at surgical excision. **(B)** Classic LCIS and ILC in a 48-year-old woman with a history of ipsilateral classic LCIS. Axial postcontrast subtraction T1-weighted image shows an interval new 0.6-cm enhancing mass (white circle) with a type 1 kinetic curve, suspicious. MRI-guided biopsy yielded classic LCIS. The lesion yielded ILC at surgical excision.

LCIS cases were associated with benign entities such as fibrocystic change and sclerosing adenosis.

On the other hand, 16 LN cases in the cohort were associated with calcifications at histopathologic analysis. Of these, 15 underwent excision, three of which (20%) were upgraded, including one case of florid LCIS and two cases of classic LCIS with necrosis. The authors suggest that the presence of nonincidental calcifications in association with LN should raise suspicion for the presence of a variant. They also suggest that calcified classic LCIS should be treated distinctly from incidental noncalcified classic LCIS, with excision being a prudent approach.

The distinction between incidental and nonincidental calcified LCIS may also explain higher upgrade rates for classic LCIS seen in some studies, even after exclusion of discordant cases and variants. One analysis reported an upgrade rate of 8.4% for classic-type LCIS even after excluding discordant cases and pleomorphic LCIS, but the cohort still included cases of calcified classic LCIS and LN manifesting as mass lesions, which may have increased the possibility of upgrade (42).

LN and MRI

There is less definitive evidence regarding how to manage LN detected at breast MRI as opposed to LN identified at mammography or US. The literature is limited, with small and retrospective studies, many of which included nonpure LN, often assessed only relatively few cases of LN as part of a larger assessment of multiple high-risk lesions combined (43), or did not perform radiologic-pathologic evaluation for discordance or distinguish between classic LCIS and the variant types (44,45). This likely contributes to the relatively high upgrade rates that have been reported for LN detected at MRI. However, even several studies that performed more nuanced pathologic assessments show high upgrade rates above a 2% threshold for which surveillance might be appropriate (46–48).

In a multi-institution study, Khoury and colleagues (47) reviewed 1655 MRI-guided biopsies, of which 63 (4%) were LN. In this group, 11 of 34 cases of pure LN (32%), including three cases of pleomorphic LCIS, were upgraded to malignancy. Even excluding the three cases of pleomorphic LCIS,

eight of 31 cases (26%) were upgraded. Of note, the authors do not comment on whether the non-pleomorphic LCIS cases included other variants, which would have implications for the upgrade rates.

Another recent study described 65 of 943 MRI-guided biopsies that resulted in LN as the highest-risk pathologic lesion at core sampling (49). In 32 of these cases, LN was the dominant finding at pathologic analysis. Of this group, one pleomorphic LCIS case and two classic LCIS cases were upgraded, for a total upgrade rate of 9.4%. The authors found no predictors of upgrade, although they observed that LN most commonly manifested as focal heterogeneous nonmass enhancement with low T2 signal intensity.

Khoury and colleagues (47) found that more of their upgraded LN cases manifested as nonmass enhancement versus mass-enhancing lesions, but this difference was not significant. More recently, Amitai and colleagues (50) found that upgrade of LN (11% upgraded cases [eight of 76]) was associated with both lesion size (mean size of 5.5 cm in upgraded cases vs 1.9 cm in nonupgraded cases) and clumped nonmass enhancement (75% vs 24%, $P = .006$). However, across the majority of studies to date, no clinical or imaging characteristics have been shown to allow statistically significant prediction of upgrade for LN identified at MRI (47,49,51) (Fig 8).

There have also been a few studies assessing whether the negative predictive power of MRI can be used to safely identify high-risk lesions that might be followed rather than excised once identified at percutaneous biopsy. Londero and colleagues (52) assessed 227 high-risk lesions and concluded that lack of contrast enhancement could allow reliable exclusion of malignancy in this setting (70 of 72 cases; negative predictive value [NPV] = 97%; 95% CI: 94, 99). However, a closer look at the 37 LN lesions with results of final excision pathologic analysis in the study shows that a high number (12 of 37 cases [32%]) were upgraded to malignancy. There were five intermediate-grade ILCs, one intermediate-grade invasive tubulolobular carcinoma, one intermediate-grade invasive ductal carcinoma (IDC), and five cases of ductal carcinoma in situ (DCIS) (two low grade, one intermediate grade, and two high grade) in the upgraded group.

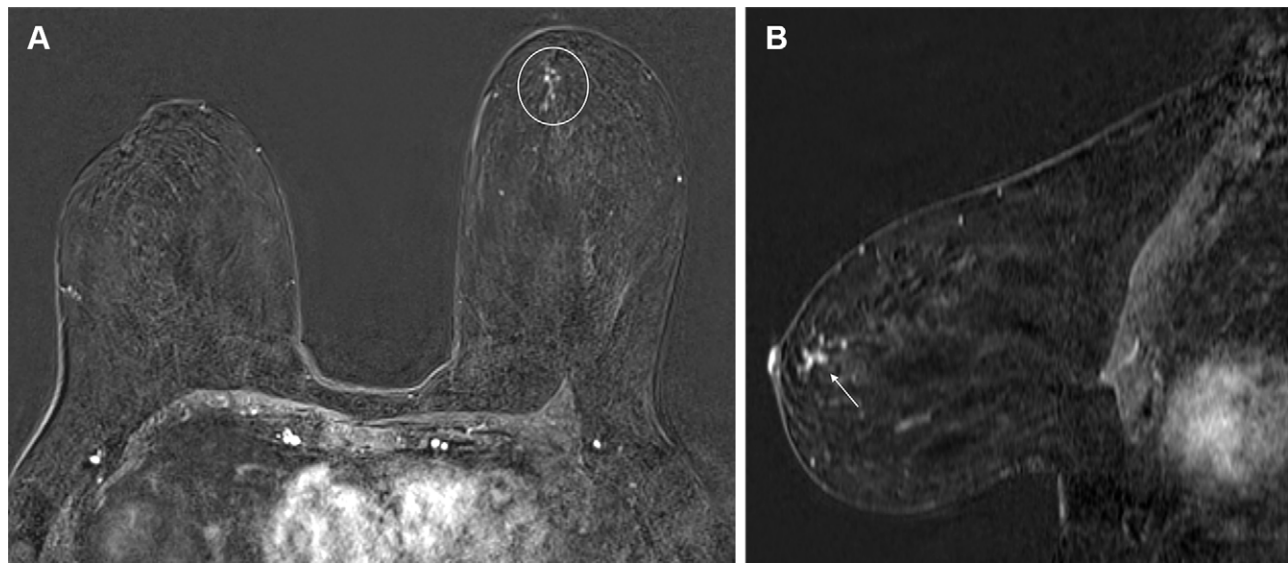


Figure 9. ILC, pleomorphic LCIS, ALH, and classic LCIS in a 65-year-old woman with a family history of premenopausal breast cancer in her mother (aged 46 years) and postmenopausal breast cancer in maternal and paternal aunts. She also had a personal history of right classic LCIS 16 years earlier and again 11 years earlier, which were treated with excision, with results of final pathologic analysis in each case yielding classic LCIS and fibrocystic changes. The patient had taken tamoxifen for 9 months but could not tolerate the medication. She presented for screening MRI. (A) Axial postcontrast subtraction T1-weighted image from annual screening MRI shows interval new 2.5-cm linear left retroareolar nonmass enhancement (white circle), a suspicious finding. Concurrent mammograms (not shown) were normal and unchanged. (B) Sagittal postcontrast subtraction T1-weighted image shows the nonmass enhancement (arrow). MRI-guided biopsy yielded pleomorphic LCIS. Surgical excision yielded foci of ILC and pleomorphic LCIS as well as ALH and classic LCIS.

The authors describe evaluation of sampling adequacy for calcifications, but any postbiopsy concordance analysis was not reported. As well, any variants in the cohort were not identified separately. There was only one false-negative case for the LN cohort. The lesion was identified at mammography as pleomorphic calcifications and demonstrated no enhancement at subsequent MRI. Low-grade DCIS was seen at excision.

Another study that assessed whether MRI could be used as a determinant of upgrade for known high-risk lesions found an “unacceptably” low NPV for upgrade of known LN at MRI (88% [22 of 25 cases]) and concluded that MRI assessment could not allow reliable determination of malignancy for this lesion (53). This suggests a need for further studies on an LN-specific cohort. At present, there are still relatively few assessments of LN upgrade rates at MRI as well as relatively few studies presented with an eye to the impact of discordance on upgrade rates. More studies are needed to help provide more nuanced management guidance.

Longer-term Management Considerations

Risk

A diagnosis of LN is associated with a greater than average lifetime risk of developing breast cancer. ALH is associated with a four to five times relative risk of developing breast cancer, and LCIS is associated with a higher eight to 10 times relative risk (8). King and colleagues (54) observed a 2% annual incidence of breast cancer in their cohort of 1004 women with a diagnosis of LN who underwent surveillance over a 29-year period (168 cancers in 150 women at a median follow-up of 81 months, range 3–368 months).

The relatively higher risk conferred by a diagnosis of LCIS compared with that conferred by a diagnosis of ALH is one argument in favor of maintaining distinct categorizations of these entities despite the potentially subjective nature of separating them out at histopathologic analysis, as discussed earlier. Surveillance, Epidemiology, and End Results (SEER) data have shown that among 19 462 women with a history of LCIS, there was a cumulative incidence of breast malignancy of 11.3% (95% CI: 10.7, 11.9) at 10 years and 19.8% (95% CI: 18.8, 20.9) at 20 years (Fig 9).

Chemoprevention

Several studies have shown the benefits of chemoprevention in women with LN. King and colleagues (54) found that breast cancer incidence was significantly reduced in women with a history of LCIS undergoing chemoprevention (10-year cumulative risk: 7% with chemoprevention, 21% with no chemoprevention; $P < .001$). National Comprehensive Cancer Network (NCCN) guidelines recommend risk-reducing endocrine therapy for individuals with a history of LCIS and a life expectancy of greater than or equal to 10 years (55).

Supplemental Screening

Several recent guidelines also recommend consideration of supplemental annual screening breast MRI in women with a history of LN. The NCCN 2022 guidelines recommend annual screening mammography to begin at diagnosis of LN (but not before age 30 years). These guidelines also suggest consideration of annual breast MRI after age 25 years for women with LN and a residual lifetime risk of greater than or equal to 20% (56). Contrast-enhanced mammography and screening US

may also be considered as alternatives for those who qualify but have a contraindication to MRI or inability to tolerate the examination.

Similarly, the 2023 American College of Radiology (ACR) screening guidelines (57) for women at higher risk of breast cancer recommend that supplemental screening MRI should be considered, in women with a history of atypia or LCIS, “especially if other risk factors are present,” an acknowledgment that risk is multifactorial and that women with a history of LN may also have other factors—dense breasts, family history—that will further compound their risk of developing later cancer.

Currently, the evidence for screening with breast MRI in the population with LN is based on observed risk in the cohort with an LN history as well as extrapolation from the highest-risk patient population literature rather than on robust imaging-based literature. There are small retrospective studies that have shown increased cancer detection rate (CDR) at MRI versus mammography in women with a history of LN (58,59). A review of supplemental MRI performance in 2637 patients with different levels of risk found that supplemental screening MRI resulted in a CDR of 15 per 1000 examinations in women with a history of LN or atypical ductal hyperplasia (ADH). Overall screening performance in women with a history of a high-risk lesion was equivalent to outcomes in *BRCA* mutation carriers without significant difference in CDR, positive predictive value (PPV), sensitivity, or specificity (60).

A study of MRI screening performance in 145 women with a history of atypia or LN also found equivalent PPV₁ and PPV₃ rates compared with those of other high-risk groups for whom MRI supplemental screening is performed (61). However, a prospective two-arm study comparing supplemental MRI to mammography alone showed no significant difference in cancer detection (62). A more recent retrospective study of 542 women with a history of ADH or ALH and 157 women with a history of LCIS who underwent either MRI and mammography or mammography alone found equivalent 4-year CDRs for breast cancer (3.6% in both groups, $P = .89$). The cancers in the two groups were all node negative and stage 0 or 1. However, there were significantly higher biopsy rates in the MRI group (30.5% vs 12.6%, hazard ratio (HR) = 2.67, $P < .001$) (63).

The mixed results in these studies likely arise from the heterogeneity of included patients, the small sample size, the overall low numbers of cancers detected in this moderate-risk population, and the relatively short follow-up periods. Recommendations for supplemental MRI screening in women with a history of LN are based more on lifetime risk observations, extrapolations from the high-risk patient literature, and recognition that women may have multiple simultaneous risk factors, rather than on robust data from the screening MRI literature.

Consensus and Guidelines

There are several broad recommendation documents by surgical societies such as the American Society of Breast Surgeons and by international bodies such as the International Consensus Conference and the National Health Service

(NHS) Breast Screening Multidisciplinary Working Group (64,65), with recommendations that vary by practice norms and tissue-sampling approaches. In the United Kingdom, for example, it is standard to manage classic LN diagnosed at 14-gauge or vacuum-assisted biopsy with percutaneous vacuum-assisted excision if the lesion is determined to be eligible for this approach after multidisciplinary team evaluation (66). Guidelines from the American Society of Breast Surgeons state that LCIS and ALH may be excised or followed, but that routine excision is no longer routinely recommended if there is no discordance and no additional high-risk lesion in the biopsy sample. However, excision should be performed if there is discordance or limited sampling or if another high-risk lesion is present. Pleomorphic LCIS should be excised, as should LCIS with necrosis and other nonclassic lesions (64).

At this time, however, there are no radiology-based national U.S. consensus guidelines for management of high-risk lesions and no specifically stated surveillance guidelines for the appropriate follow-up period and Breast Imaging Reporting and Data System (BI-RADS) categorization that should be used to follow high-risk lesions. While short-term interval surveillance protocols following a BI-RADS 3 model with 6-month, 12-month, and 24-month surveillance for ALH or classic LCIS or both have been suggested (42), there remains a lack of national guidelines and consensus on how surveillance should be managed.

Conclusion

Appropriate management decisions after LN is encountered at biopsy demand understanding of the heterogeneity of LN, recognition of the LCIS variants, and careful radiologic-pathologic assessment and evaluation of concordance. For ALH, there is ample evidence to support surveillance if the biopsy results are concordant and there are no additional associated lesions that would warrant excision. Incidental noncalcified classic LCIS also shows very low upgrade rates, and surveillance is appropriate when scrupulous radiologic-pathologic correlation is practiced.

There is still a need for consensus and national U.S. radiology-based guidelines both overall and specifically in terms of surveillance intervals and duration of surveillance follow-up. At this time, there is a need for more studies on the optimal approach for LN identified at MRI. For all modalities, if there is any uncertainty about radiologic-pathologic correlation, excision will be the most prudent option. Finally, both ALH and LCIS confer a greater than average lifetime risk of later development of breast cancer, warranting referral of patients for discussion of long-term risk management strategies.

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